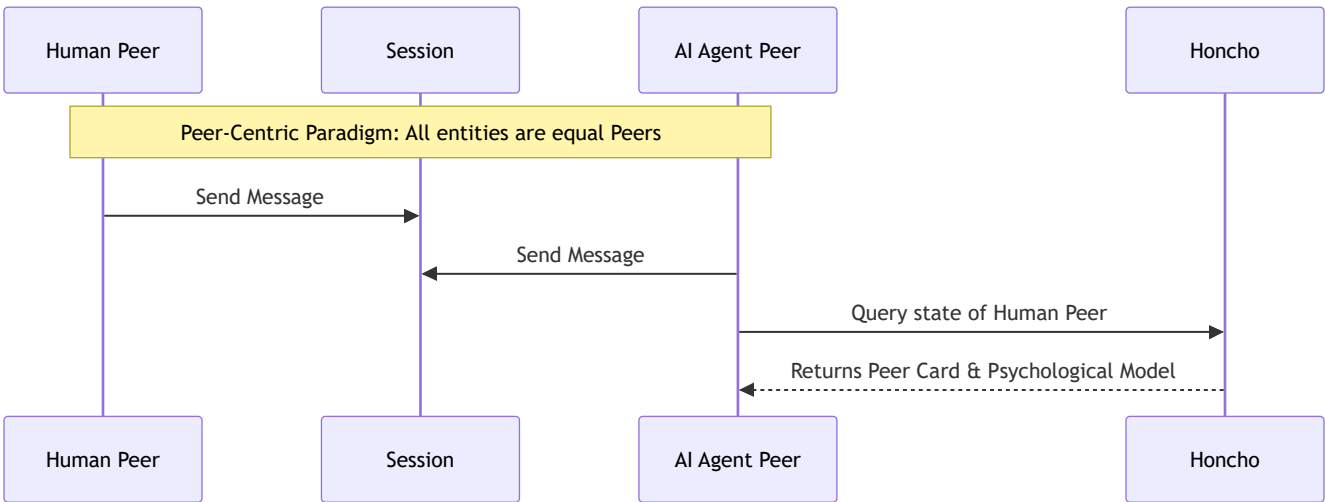


Research Report: Honcho Memory Service Architecture

1. Philosophical Foundations

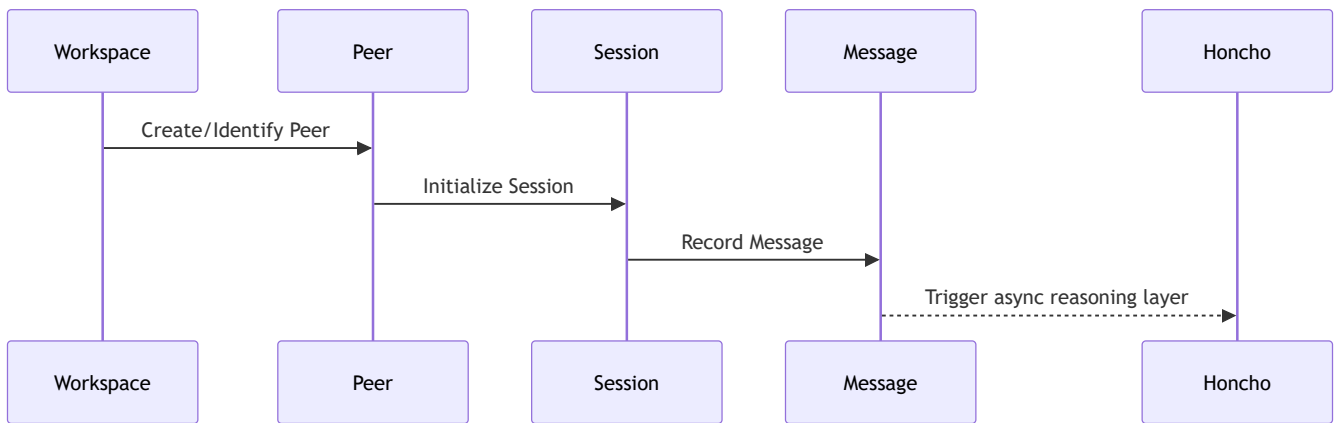
Honcho distinguishes itself from traditional passive Retrieval-Augmented Generation (RAG) by championing an **active, reasoning-driven memory** approach. Its goal is not mere fact retrieval, but true "stateful simulation" achieved by constructing coherent psychological models of entities.

- **Formal Logic vs. Plausible Text:** Honcho leverages custom language models trained in formal logic to extract latent knowledge and draw deductive conclusions rather than just surfacing similar text chunks.
- **The Peer-Centric Paradigm:** Honcho abandons the traditional "user vs. assistant" dynamic. All entities—humans, agents, or groups—are equally treated as "Peers". This allows multi-agent systems to model and reason about other agents exactly as they do human users.



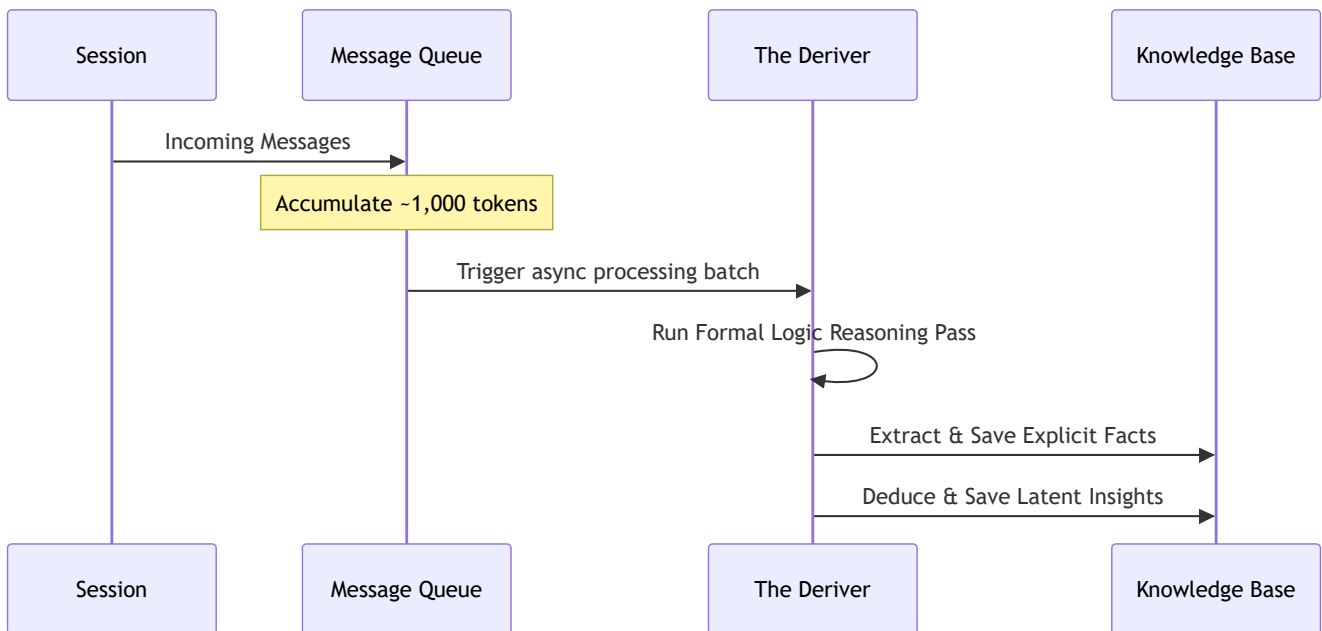
2. Core Architecture

The system employs a hierarchical data model (Workspaces -> Peers -> Sessions -> Messages) and splits operations into a standard memory storage layer and a background Reasoning Layer. The Reasoning Layer consists of three key agents:



The Deriver (Ingestion & Extraction)

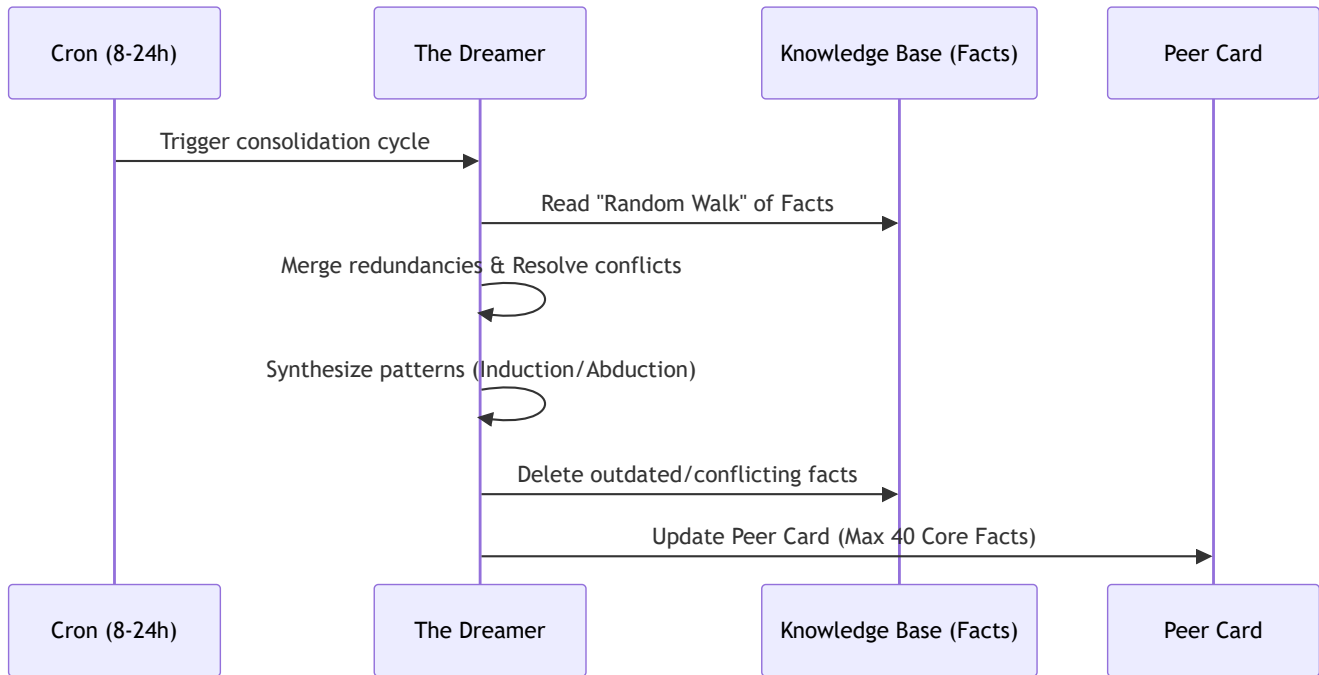
- Acts as the immediate reasoning engine for incoming data.
- Processes messages asynchronously via a queue.
- Uses **token batching** (triggering inference roughly every 1,000 tokens) to ensure meaningful context while keeping API costs low.
- Extracts explicit facts and derives unstated deductive insights.



The Dreamer (Consolidation)

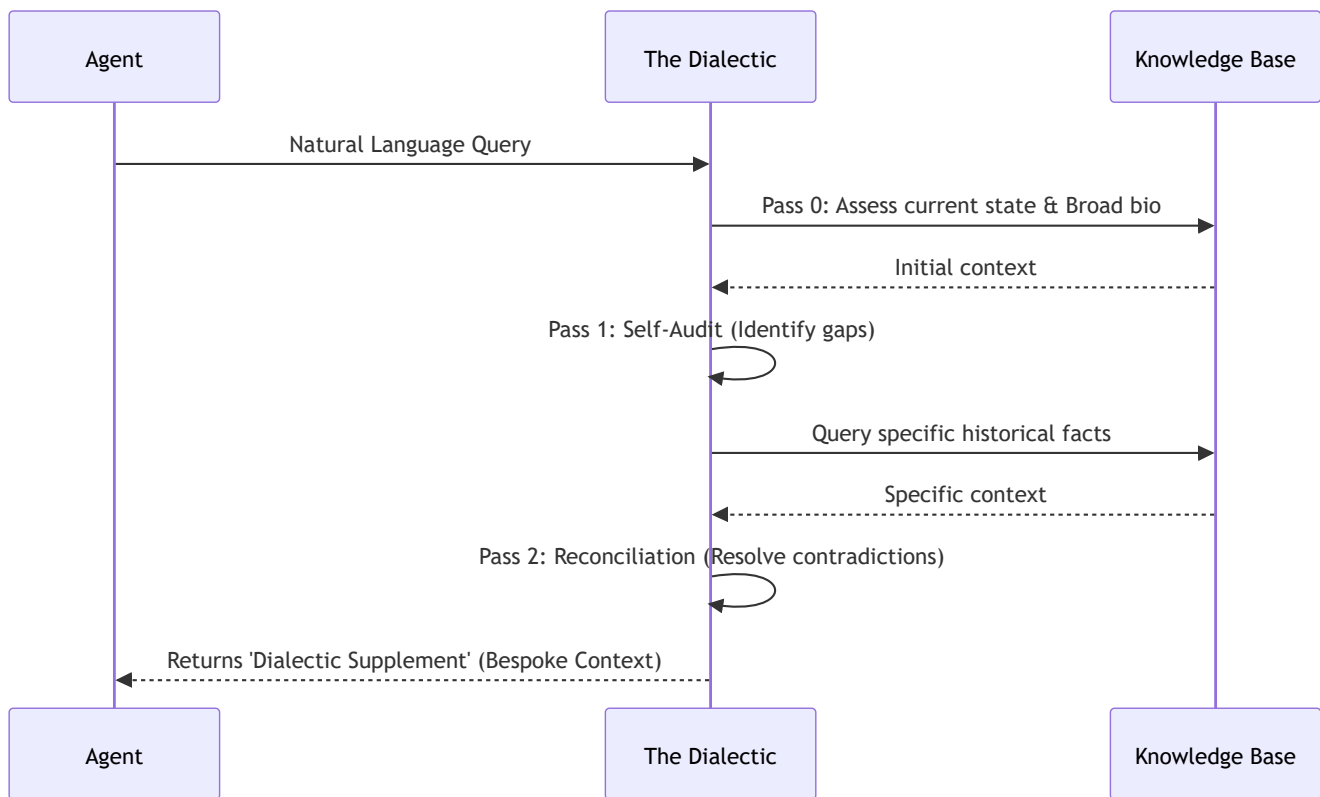
- A background maintenance agent running every 8 to 24 hours.
- Performs a "random walk" across peer observations to consolidate memory: merging redundancies, deleting outdated or conflicting information, and synthesizing specific facts into broader patterns (induction and abduction).

- Its primary output is the **Peer Card**, a highly compressed biographical profile (hard-capped at 40 facts) that is injected into the agent's prompt to bypass retrieval latency.



The Dialectic (Retrieval & Synthesis)

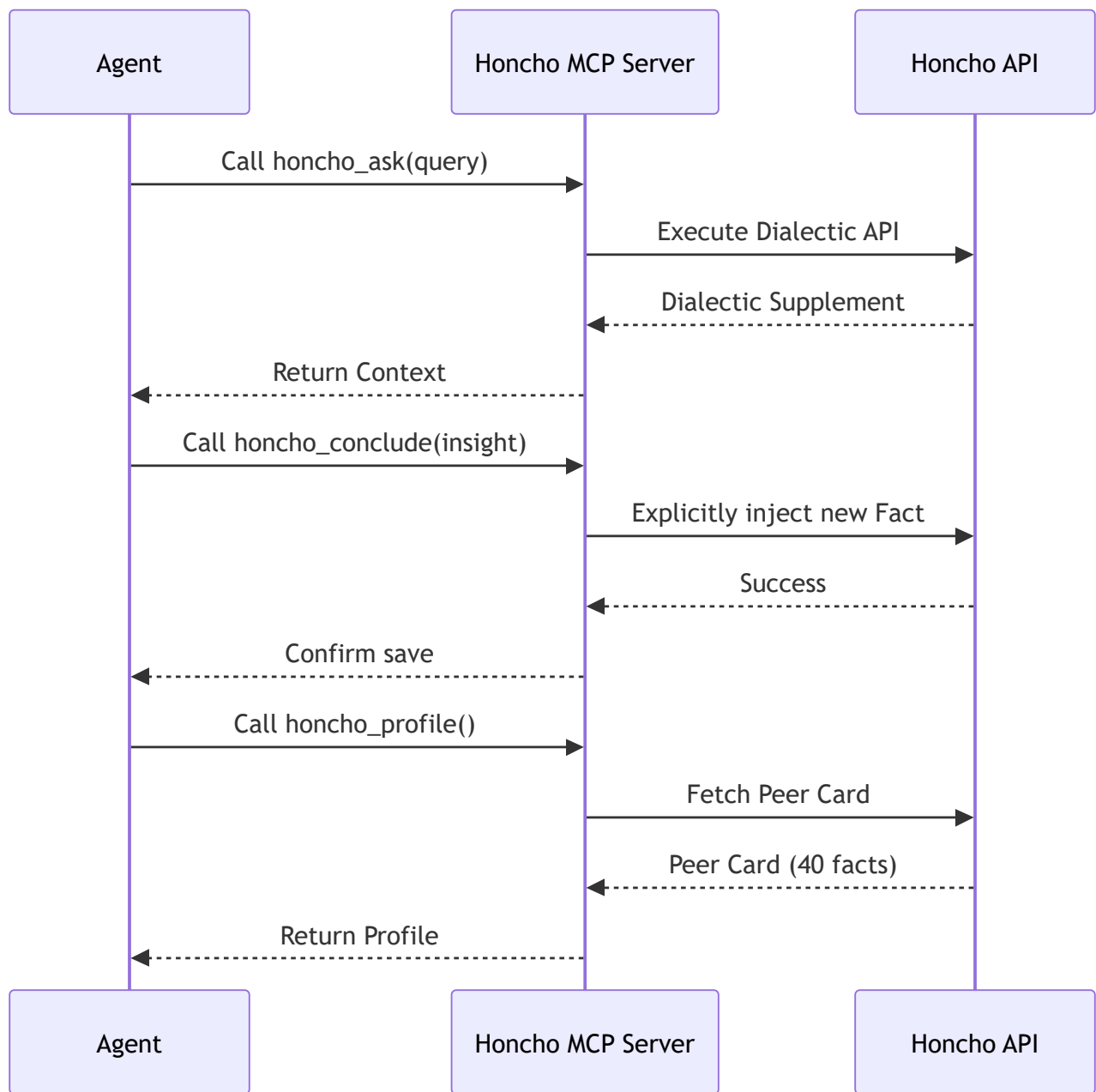
- A natural language retrieval API that acts as an "Oracle" for querying peer memory.
- Uses **Multi-Pass Reasoning**: Pass 0 (Assessment), Pass 1 (Self-Audit), and Pass 2 (Reconciliation of contradictions).
- Automatically toggles between "Cold Start" (broad biography) and "Warm Session" (scoped to recent context).
- Outputs a **Dialectic Supplement**—real-time LLM-synthesized reasoning about the user's current needs—which is injected alongside the base context on every conversational turn.



3. Honcho MCP Tool Integrations

The Honcho Model Context Protocol (MCP) server provides agents with direct manipulation capabilities over the memory layer:

- `honcho_context` : Retrieves the full cross-session user representation (summaries, peer cards, relevant observations).
- `honcho_ask` : An LLM-powered Q&A tool leveraging the Dialectic API. Supports configurable reasoning depths (quick vs. thorough).
- `honcho_conclude` : Allows the agent to explicitly save a new insight or fact as a "conclusion".
- `honcho_profile` : Retrieves or updates the user's biographical "peer card".
- `honcho_search_conclusions` : Performs semantic search over the derived insights for high-fidelity fact recall.
- `honcho_search_messages` : Queries historical session messages (filterable by date and sender).
- `get_config` / `set_config` : Utility tools to programmatically inspect or modify memory configurations.



4. Visual Diagrams

4.1 Comprehensive Honcho Master Pipeline

This diagram visualizes how the Philosophical Foundations (Peer-Centric), Data Architecture, Active Reasoning Engines (Deriver, Dreamer, Dialectic), and the MCP Tool Integrations all connect into a single unified flow.

